Engineering Report – AeroGrid

Analysis of the 24-hour telemetry dataset identified two turbines exceeding safe operating thresholds. Turbine T-04 recorded an average temperature of 90.58°C, exceeding the 85°C limit and indicating sustained overheating. Turbine T-07 recorded a maximum vibration level of 25.0 mm/s, significantly above the 15.0 mm/s threshold, suggesting abnormal mechanical behaviour and potential component failure. Both turbines should be prioritised for inspection and maintenance.

To address the limitations of the current single-server system, a cloud-based streaming architecture has been proposed. Telemetry data is first received by AWS IoT Core, which provides a scalable and secure ingestion layer for IoT devices. Data is then sent to AWS Kinesis, which acts as a message queue, buffering incoming telemetry and preventing data loss during periods of high traffic. This removes the risk of sensors overwhelming downstream systems.

AWS Lambda processes the telemetry stream in real time, applying anomaly detection rules as data arrives. This enables immediate identification of overheating or excessive vibration rather than relying on delayed batch analysis. Recent telemetry is stored in DynamoDB, providing fast access for dashboards and operational monitoring. Historical data is archived to Amazon S3, a low-cost data lake suitable for long-term storage and future analytical workloads.

This architecture eliminates the single point of failure present in the legacy system by separating ingestion, processing and storage into independent, scalable services.

To optimise costs, telemetry data should be aggregated into one-minute intervals before long-term storage. This reduces storage requirements and processing costs while retaining sufficient detail for trend analysis, reporting and anomaly detection.